

Heteroscedastic mixed-variable BO benchmark — progress report

July 22, 2026

1 Scope and status of the runs

The benchmark compares **4 surrogate models** (Separate GP, Categorical-kernel GP, Standard LVGP, Heteroscedastic LVGP) $\times$ **6 acquisitions** (EI/LCB/PI + noise-aware HAEI/ANPEI/RAHBO) $\times n_{\text{rep}} \in \{3, 10\}$ replicates $\times 5$ seeds on 9 mixed-variable heteroscedastic test problems (200 BO iterations for the multi-dimensional problems, 50 for the 1-D ones; shared SLHD initial designs per seed — common random numbers across all models). The sweep runs python models first, all $n_{\text{rep}}=10$ blocks before $n_{\text{rep}}=3$, and the 10-D problem last.

Problem	Complete	Note
ackley_2d, branin_hetero, griewank_2d, sixhump_camel	100%	1-D; also carry the full 30-seed evidence from the earlier sweep
otl_circuit, piston, rastrigin_6d	100%	multi-dimensional (d = 5–6)
golinski	97%	6 cells lost to an early 8 h timeout (since raised to 16 h); refill pass pending
ackley_10d	32%	<i>in progress</i> — heter-LVGP cells still pending

Table 1: Run status as of July 22, 2026 (sweep position 1748/2100 grid cells).

2 Computational cost and the 5-seed decision

	Separate GP	Categorical kernel	Standard LVGP	Heteroscedastic LVGP
1-D problems (50 iter)	18–44 s	54–114 s	5–11 min	5–13 min
d = 5–6 problems (200 iter)	20–40 min	43–110 min	4.3–7.3 h	4.6–6.9 h
ackley_10d (200 iter)	1.1 h	3.2 h	20.1 h	pending (projected ~ 1 –1.5 d)

Table 2: Measured mean wall-clock per cell (one full BO run). The MATLAB LVGP engines cost ~ 10 – $25\times$ the python GP models; the validated native python LVGP port (`compare_matlab_python.ipynb`) is the mitigation path for the 10-D block.

Why 5 seeds. The sweep runs 24 parallel workers on the shared node, so the constraint is total compute, not sequential time: at the measured per-cell costs the full 30-seed grid amounts to roughly 3×10^4 core-hours (dominated by the MATLAB LVGPs on the $d \geq 5$ problems and the 10-D block), i.e. $\approx 7\text{--}8$ *weeks* of wall-clock even at full 24-way parallelism — monopolizing a node shared with other users. Five seeds with both n_{rep} cut this to ≈ 9 days of wall-clock (consistent with the observed pace: $\sim 85\%$ of the grid in 6 days) while still populating *every* (model, acquisition, problem) cell, seed-paired across models via the shared initial designs. The analysis compensates statistically (medians/IQR, cross-problem rank consistency rather than per-problem significance), the four 1-D problems retain their 30-seed evidence, and the harness is resumable so seeds can be extended selectively later.

3 Regret convergence

regret — EI, $n_{\text{rep}}=10$, incumbent by true f

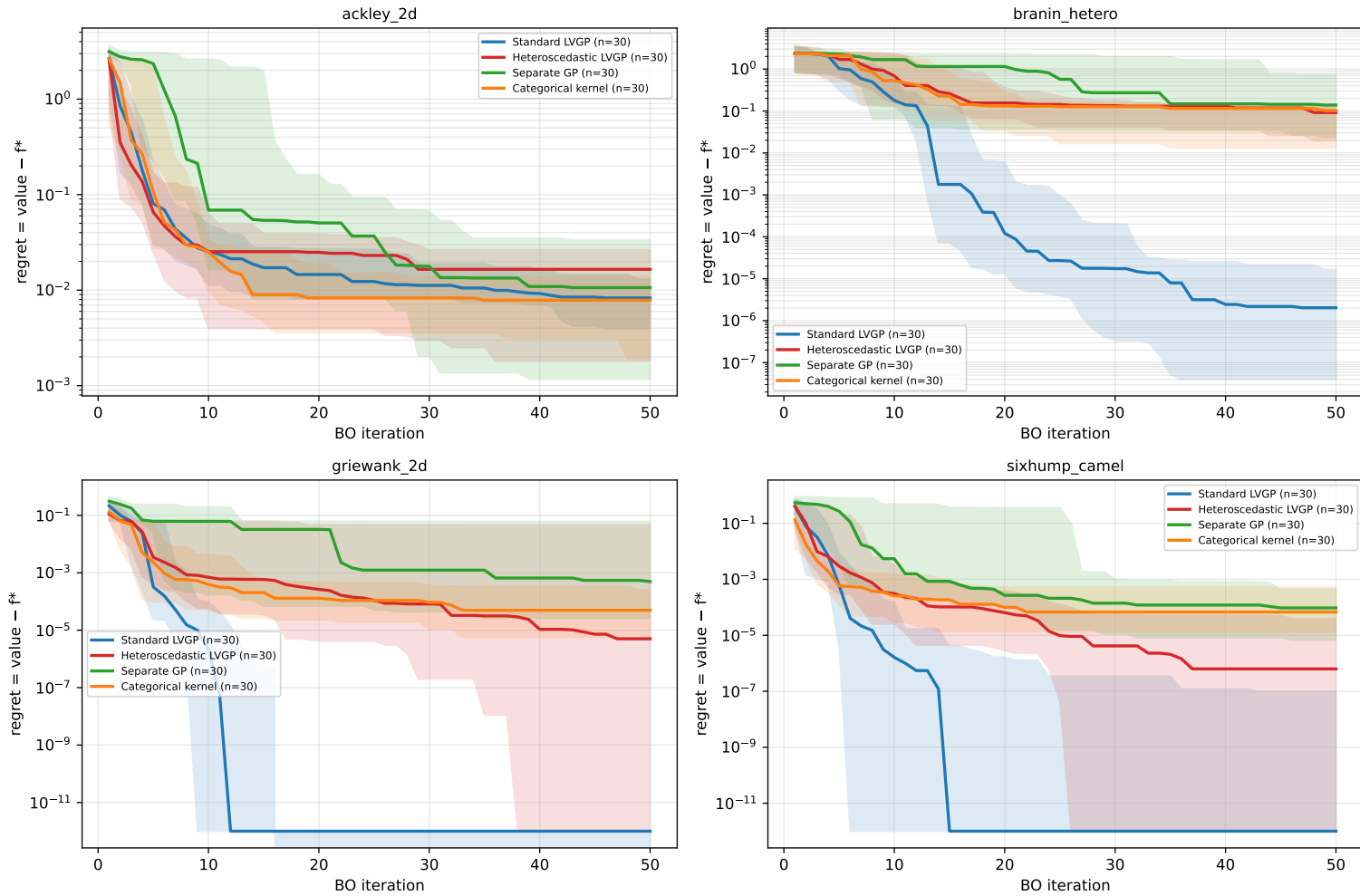


Figure 1: Regret convergence (EI, $n_{\text{rep}} = 10$, median $\pm$ IQR over seeds) — 1-D problems. Other acquisitions: `compare.ipynb` Part 3/6.

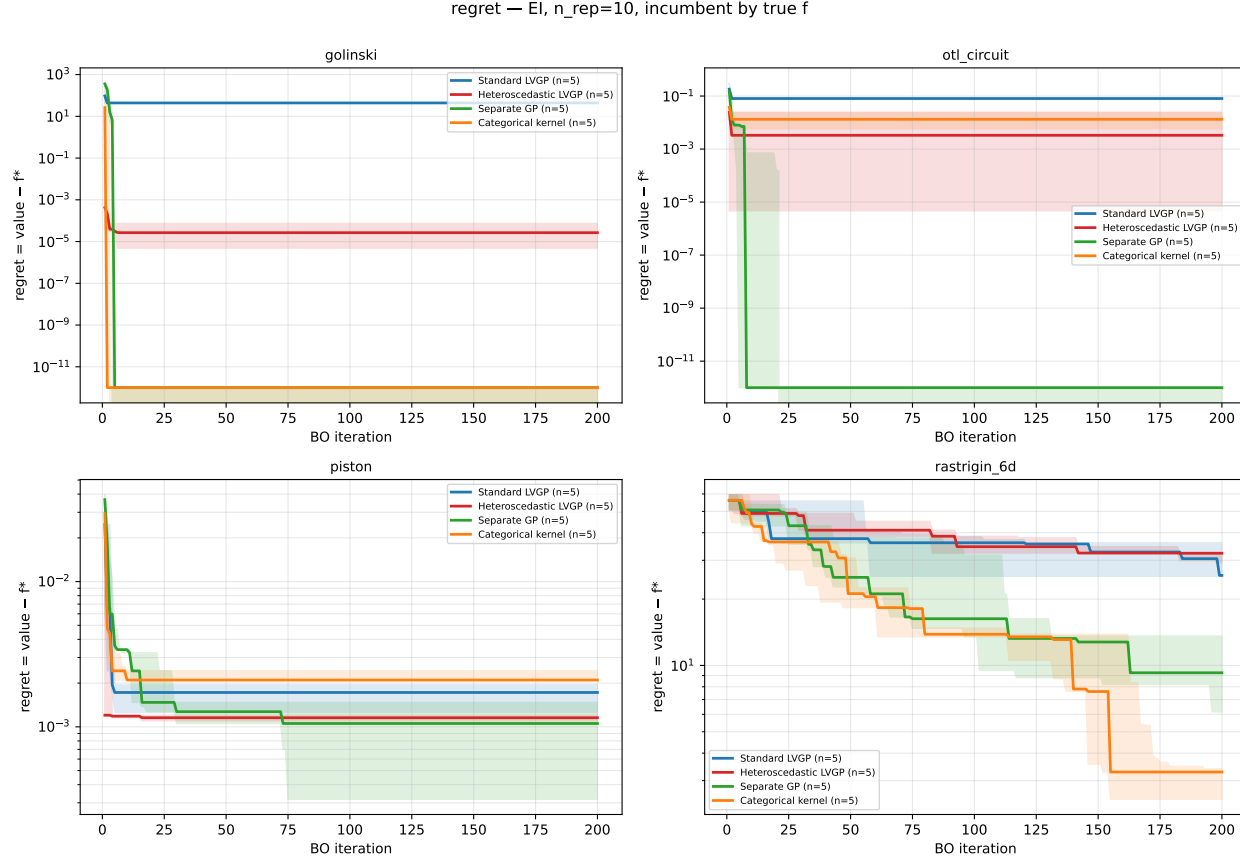


Figure 2: Regret convergence (EI, $n_{\text{rep}} = 10$) — multi-dimensional problems.

Findings. (1) On the 1-D problems every model converges to the optimum region; the MATLAB LVGPs exploit it tightest — note this partly reflects their BO loop, not only the model (verified: the loop effect appears for non-LVGP models too), so cross-engine regret gaps should be read with care. (2) **golinski exposes noise-blindness:** the Standard LVGP’s median final regret under EI is ≈ 44 (PI: ≈ 49) while the heteroscedastic LVGP and both python GPs solve the problem to ≈ 0 — the optimum sits in a region with replicate variance $\sigma^2 = 576$, and fitting replicate means as if noiseless breaks the model there. (3) **rastrigin_6d is the hardest problem so far** and is led by the python GPs (median EI regret: categorical 3.3, separate 9.3) with both LVGPs trailing (standard 25.6, heter 32.3) — surface accuracy dominates on this rugged landscape. (4) *otl_circuit* and *piston* are near-solved by all models (regret ≤ 0.08); differences there are in *design noise*, not in f (see §5-6).

4 Final regret tables

Legend (all tables). Cell = final metric (mean $\pm$ sd over seeds). **Bold** = best model for that acquisition; **green** = best (acquisition, model) combination for that problem. Lower is better.

Table 3: Final regret (value $- f^*$) for every acquisition $\times$ surrogate model ($n_{\text{rep}} = 10$, mean $\pm$ sd over seeds).

Acquisition	Model	Ackley 2-D	Branin	Griewank 2-D	Six-hump Camel
LCB	Standard LVGP	0.00408 $\pm$ 0.0047	0.00124 $\pm$ 0.0059	0.00413 $\pm$ 0.015	0.0322 $\pm$ 0.12
	Heteroscedastic LVGP	0.0188 $\pm$ 0.021	0.0993 $\pm$ 0.063	0.0105 $\pm$ 0.023	0.0323 $\pm$ 0.12
	Separate GP	0.726 $\pm$ 1.3	0.529 $\pm$ 0.88	0.0769 $\pm$ 0.094	0.2 $\pm$ 0.33
	Categorical kernel	0.294 $\pm$ 0.88	0.0941 $\pm$ 0.12	0.0183 $\pm$ 0.06	0.0185 $\pm$ 0.098
PI	Standard LVGP	0.0131 $\pm$ 0.014	0.00462 $\pm$ 0.023	0.0009 $\pm$ 0.0012	0.000114 $\pm$ 0.00017
	Heteroscedastic LVGP	0.28 $\pm$ 1	0.238 $\pm$ 0.59	0.0426 $\pm$ 0.081	0.0488 $\pm$ 0.15
	Separate GP	0.709 $\pm$ 1.3	0.238 $\pm$ 0.59	0.0349 $\pm$ 0.062	0.0684 $\pm$ 0.19
	Categorical kernel	0.00491 $\pm$ 0.006	0.0548 $\pm$ 0.066	0.0025 $\pm$ 0.011	0.000646 $\pm$ 0.0023
EI	Standard LVGP	0.0102 $\pm$ 0.0098	0.00181 $\pm$ 0.0076	$-6.54 \times 10^{-7} \pm 4.2 \times 10^{-7}$	$1.81 \times 10^{-6} \pm 5.8 \times 10^{-6}$
	Heteroscedastic LVGP	0.022 $\pm$ 0.031	0.0802 $\pm$ 0.058	0.0225 $\pm$ 0.048	0.0324 $\pm$ 0.12
	Separate GP	0.422 $\pm$ 1	0.662 $\pm$ 0.95	0.029 $\pm$ 0.049	0.0457 $\pm$ 0.18
	Categorical kernel	0.01 $\pm$ 0.0092	0.158 $\pm$ 0.39	0.000174 $\pm$ 0.00025	0.000521 $\pm$ 0.00094
HAEI ($\gamma = 0.5$)	Heteroscedastic LVGP	0.0259 $\pm$ 0.023	0.0211 $\pm$ 0.035	0.00868 $\pm$ 0.021	0.0163 $\pm$ 0.087
	Separate GP	0.425 $\pm$ 1	0.434 $\pm$ 0.74	0.0288 $\pm$ 0.049	0.0486 $\pm$ 0.15
	Categorical kernel	0.00801 $\pm$ 0.0085	0.0421 $\pm$ 0.048	0.0095 $\pm$ 0.021	0.000346 $\pm$ 0.00085
ANPEI ($\beta = 0.8$)	Heteroscedastic LVGP	0.0208 $\pm$ 0.019	0.00541 $\pm$ 0.014	0.0194 $\pm$ 0.028	$6.35 \times 10^{-5} \pm 0.00015$
	Separate GP	0.116 $\pm$ 0.55	0.666 $\pm$ 1.1	0.0413 $\pm$ 0.048	0.000237 $\pm$ 0.00041
	Categorical kernel	0.00552 $\pm$ 0.0094	0.135 $\pm$ 0.41	0.0406 $\pm$ 0.049	0.000346 $\pm$ 0.00089
RAHBO ($\alpha = 0.5$)	Heteroscedastic LVGP	0.0203 $\pm$ 0.025	0.00189 $\pm$ 0.0049	0.0085 $\pm$ 0.021	0.0162 $\pm$ 0.087
	Separate GP	0.723 $\pm$ 1.3	0.288 $\pm$ 0.78	0.0768 $\pm$ 0.094	0.168 $\pm$ 0.3
	Categorical kernel	0.397 $\pm$ 1	0.154 $\pm$ 0.57	0.0411 $\pm$ 0.1	0.000219 $\pm$ 0.00055

Table 4: Final regret (value $-f^*$) for every acquisition $\times$ surrogate model ($n_{\text{rep}} = 10$, mean $\pm$ sd over seeds).

Acquisition	Model	Golinski	OTL Circuit	Piston	Rastrigin 6-D
LCB	Standard LVGP	$2.5 \times 10^{-6} \pm 8.5 \times 10^{-8}$	$3.87 \times 10^{-6} \pm 3.2 \times 10^{-6}$	$2.16 \times 10^{-7} \pm 1.5 \times 10^{-7}$	28.3 ± 6.1
	Heteroscedastic LVGP	$1.26 \times 10^{-7} \pm 1.3 \times 10^{-9}$	$4.64 \times 10^{-7} \pm 3.8 \times 10^{-7}$	$4.62 \times 10^{-7} \pm 5.1 \times 10^{-7}$	25 ± 5.7
	Separate GP	0 ± 0	0.00142 ± 0.0028	0.00104 ± 0.00071	18.2 ± 9.6
	Categorical kernel	0 ± 0	0 ± 0	0.00105 ± 0.00023	6.49 ± 4.2
PI	Standard LVGP	43.8 ± 0	0.055 ± 0.03	0.00127 ± 0.00052	20.7 ± 6.8
	Heteroscedastic LVGP	1.37 ± 2.6	0.00558 ± 0.0096	$2.93 \times 10^{-5} \pm 3.7 \times 10^{-5}$	23.4 ± 3.2
	Separate GP	0 ± 0	$6.94 \times 10^{-5} \pm 7 \times 10^{-5}$	0.000532 ± 0.00038	13.6 ± 3.8
	Categorical kernel	0 ± 0	0.0158 ± 0.011	0.00125 ± 0.0011	3.87 ± 2.8
EI	Standard LVGP	37.3 ± 13	0.0659 ± 0.03	0.0025 ± 0.0021	30.6 ± 9.4
	Heteroscedastic LVGP	$6.39 \times 10^{-5} \pm 7.9 \times 10^{-5}$	0.0121 ± 0.014	0.00109 ± 0.0006	31.2 ± 2.5
	Separate GP	0 ± 0	$3.07 \times 10^{-5} \pm 6.1 \times 10^{-5}$	0.001 ± 0.00074	12.2 ± 8
	Categorical kernel	0 ± 0	0.0163 ± 0.014	0.00205 ± 0.00054	3.89 ± 2.2
HAEI ($\gamma = 0.5$)	Heteroscedastic LVGP	7.82 ± 4.1	0.0117 ± 0.013	0.00336 ± 0.0018	30.4 ± 4.9
	Separate GP	0 ± 0	$3.07 \times 10^{-5} \pm 6.1 \times 10^{-5}$	0.000822 ± 0.00057	9.46 ± 2.4
	Categorical kernel	0 ± 0	0.0155 ± 0.013	0.0026 ± 0.00096	4.8 ± 1.6
ANPEI ($\beta = 0.8$)	Heteroscedastic LVGP	0.965 ± 1.4	0.0039 ± 0.0073	0.00173 ± 0.0013	24.3 ± 5.9
	Separate GP	0 ± 0	0.000162 ± 0.00029	0.00209 ± 0.0007	12.6 ± 3.4
	Categorical kernel	0 ± 0	0.0287 ± 0.017	0.00238 ± 0.0008	3.95 ± 2
RAHBO ($\alpha = 0.5$)	Heteroscedastic LVGP	2.44 ± 2.5	$2.74 \times 10^{-7} \pm 1.7 \times 10^{-7}$	$4.58 \times 10^{-7} \pm 5.5 \times 10^{-7}$	20.5 ± 4.5
	Separate GP	17.5 ± 28	0.00142 ± 0.0028	0.00123 ± 0.00062	20.4 ± 7.4
	Categorical kernel	10.6 ± 4	0 ± 0	0.000715 ± 0.00033	5.19 ± 2.6

Table 5: Final regret (value $-f^*$) for every acquisition $\times$ surrogate model ($n_{\text{rep}} = 3$, mean $\pm$ sd over seeds).

Acquisition	Model	Ackley 2-D	Branin	Griewank 2-D	Six-hump Camel
LCB	Standard LVGP	0.00641 $\pm$ 0.0063	0.0911 $\pm$ 0.057	0.00413 $\pm$ 0.015	0.0322 $\pm$ 0.12
	Heteroscedastic LVGP	0.0302 $\pm$ 0.035	0.0913 $\pm$ 0.057	0.00858 $\pm$ 0.021	0.0326 $\pm$ 0.12
	Separate GP	0.624 $\pm$ 1.2	0.649 $\pm$ 1.1	0.0912 $\pm$ 0.1	0.172 $\pm$ 0.33
	Categorical kernel	0.196 $\pm$ 0.71	0.0516 $\pm$ 0.088	0.00441 $\pm$ 0.015	0.0168 $\pm$ 0.087
PI	Standard LVGP	0.0127 $\pm$ 0.015	0.0697 $\pm$ 0.057	0.000736 $\pm$ 0.0011	0.000135 $\pm$ 0.00025
	Heteroscedastic LVGP	0.547 $\pm$ 1.2	0.289 $\pm$ 0.91	0.0395 $\pm$ 0.072	0.0345 $\pm$ 0.12
	Separate GP	0.12 $\pm$ 0.55	0.179 $\pm$ 0.42	0.026 $\pm$ 0.048	0.06 $\pm$ 0.18
	Categorical kernel	0.0145 $\pm$ 0.012	0.0736 $\pm$ 0.088	0.000795 $\pm$ 0.0019	0.00107 $\pm$ 0.0013
EI	Standard LVGP	0.0123 $\pm$ 0.012	0.0532 $\pm$ 0.054	3.11 $\times 10^{-7} \pm 3.7 \times 10^{-6}$	7.92 $\times 10^{-6} \pm 3.3 \times 10^{-5}$
	Heteroscedastic LVGP	0.0212 $\pm$ 0.024	0.092 $\pm$ 0.049	0.00697 $\pm$ 0.019	0.0369 $\pm$ 0.12
	Separate GP	0.221 $\pm$ 0.76	0.419 $\pm$ 0.71	0.0292 $\pm$ 0.049	0.0326 $\pm$ 0.12
	Categorical kernel	0.0132 $\pm$ 0.014	0.0898 $\pm$ 0.092	0.000435 $\pm$ 0.00059	0.000441 $\pm$ 0.00073
HAEI ($\gamma = 0.5$)	Heteroscedastic LVGP	0.0166 $\pm$ 0.016	0.0862 $\pm$ 0.054	0.00464 $\pm$ 0.016	0.000791 $\pm$ 0.0013
	Separate GP	0.217 $\pm$ 0.77	0.0966 $\pm$ 0.12	0.0271 $\pm$ 0.049	0.0462 $\pm$ 0.18
	Categorical kernel	0.0122 $\pm$ 0.012	0.0842 $\pm$ 0.088	0.00088 $\pm$ 0.0015	0.000906 $\pm$ 0.0011
ANPEI ($\beta = 0.8$)	Heteroscedastic LVGP	0.024 $\pm$ 0.025	0.078 $\pm$ 0.085	0.0137 $\pm$ 0.024	0.000224 $\pm$ 0.00067
	Separate GP	0.027 $\pm$ 0.027	0.19 $\pm$ 0.39	0.0467 $\pm$ 0.061	0.0341 $\pm$ 0.12
	Categorical kernel	0.0191 $\pm$ 0.024	0.131 $\pm$ 0.41	0.0267 $\pm$ 0.029	0.00284 $\pm$ 0.0053
RAHBO ($\alpha = 0.5$)	Heteroscedastic LVGP	0.0264 $\pm$ 0.024	0.00249 $\pm$ 0.0073	0.00853 $\pm$ 0.021	0.017 $\pm$ 0.087
	Separate GP	0.723 $\pm$ 1.3	0.377 $\pm$ 0.91	0.1 $\pm$ 0.12	0.188 $\pm$ 0.33
	Categorical kernel	0.11 $\pm$ 0.55	0.0269 $\pm$ 0.081	0.00463 $\pm$ 0.015	0.000752 $\pm$ 0.002

Table 6: Final regret (value $- f^*$) for every acquisition $\times$ surrogate model ($n_{\text{rep}} = 3$, mean $\pm$ sd over seeds).

Acquisition	Model	Golinski	OTL Circuit	Piston	Rastrigin 6-D
LCB	Standard LVGP	$2.56 \times 10^{-6} \pm 8.5 \times 10^{-8}$	$1.53 \times 10^{-6} \pm 9.7 \times 10^{-7}$	0.000307 ± 0.00029	23.5 ± 8.7
	Heteroscedastic LVGP	$1.24 \times 10^{-7} \pm 1 \times 10^{-8}$	$4.33 \times 10^{-7} \pm 5.5 \times 10^{-7}$	$1.49 \times 10^{-6} \pm 1.5 \times 10^{-6}$	22.7 ± 6.9
	Separate GP	0 ± 0	0.00142 ± 0.0028	0.0011 ± 0.0007	18.9 ± 8.3
	Categorical kernel	0 ± 0	$3.07 \times 10^{-5} \pm 6.1 \times 10^{-5}$	0.000715 ± 0.00033	7.57 ± 4.4
PI	Standard LVGP	31.6 ± 17	0.0242 ± 0.022	0.00168 ± 0.001	11.6 ± 3.7
	Heteroscedastic LVGP	0.941 ± 1.9	0.00164 ± 0.0012	0.000361 ± 0.0004	19.1 ± 2.4
	Separate GP	0 ± 0	$1.74 \times 10^{-5} \pm 2.3 \times 10^{-5}$	0.000718 ± 0.00046	17.2 ± 6.5
	Categorical kernel	0 ± 0	0.0155 ± 0.011	0.00177 ± 0.00093	4.18 ± 3.2
EI	Standard LVGP	16.3 ± 17	0.0565 ± 0.02	0.00339 ± 0.0033	19.5 ± 12
	Heteroscedastic LVGP	0.000119 ± 0.00015	0.0119 ± 0.013	0.00182 ± 0.0015	24.4 ± 8.1
	Separate GP	0 ± 0	0.00033 ± 0.00041	0.00131 ± 0.00074	15.1 ± 8.4
	Categorical kernel	0 ± 0	0.0218 ± 0.018	0.00267 ± 0.00088	5.61 ± 3.9
HAEI ($\gamma = 0.5$)	Heteroscedastic LVGP	3.75 ± 2.1	0.0268 ± 0.029	0.00197 ± 0.0013	28.1 ± 8.4
	Separate GP	0 ± 0	0.000173 ± 0.00028	0.00163 ± 0.00073	9.95 ± 3.9
	Categorical kernel	0 ± 0	0.0203 ± 0.017	0.00189 ± 0.00073	4.92 ± 1.7
ANPEI ($\beta = 0.8$)	Heteroscedastic LVGP	0.307 ± 0.61	0.0147 ± 0.012	0.00246 ± 0.0012	26.3 ± 5.6
	Separate GP	0 ± 0	0.000322 ± 0.0004	0.00254 ± 0.00099	11.2 ± 4.1
	Categorical kernel	1.49 ± 3	0.0146 ± 0.013	0.00295 ± 0.0018	7.34 ± 5.3
RAHBO ($\alpha = 0.5$)	Heteroscedastic LVGP	9.65 ± 6.6	$6.33 \times 10^{-7} \pm 4.8 \times 10^{-7}$	$1.71 \times 10^{-6} \pm 3 \times 10^{-6}$	22.4 ± 8.3
	Separate GP	56.7 ± 47	0.00142 ± 0.0028	0.00112 ± 0.00073	20.1 ± 10
	Categorical kernel	20 ± 29	$1.23 \times 10^{-5} \pm 2.5 \times 10^{-5}$	0.000713 ± 0.00054	8.7 ± 4.2

Findings. (1) $n_{\text{rep}} = 10$ vs 3: more replicates help the noise-aware configurations most (their noise estimates stabilize); the ranking of models is broadly stable across both. (2) The best single configurations per problem (green cells) are spread across models — no single (model, acquisition) wins everywhere; the heteroscedastic LVGP collects its wins on the problems where noise matters (golinski, otl, piston), the categorical GP on the rugged rastrigin.

5 Noise at the incumbent (lower = quieter design)

Table 7: Final noise σ^2 at the incumbent design for every acquisition $\times$ surrogate model ($n_{\text{rep}} = 10$, mean $\pm$ sd over seeds).

Acquisition	Model	Ackley 2-D	Branin	Griewank 2-D	Six-hump Camel
LCB	Standard LVGP	$0.01 \pm 2.6 \times 10^{-8}$	1.41 ± 0.0058	0.00608 ± 0.0012	0.00731 ± 0.0062
	Heteroscedastic LVGP	$0.01 \pm 4.1 \times 10^{-7}$	27.9 ± 22	0.0056 ± 0.0018	0.00731 ± 0.0062
	Separate GP	0.017 ± 0.013	22.8 ± 24	0.00788 ± 0.0068	0.0121 ± 0.013
	Categorical kernel	0.0121 ± 0.0075	20.2 ± 22	0.00712 ± 0.0034	0.00648 ± 0.0045
PI	Standard LVGP	$0.01 \pm 1.9 \times 10^{-7}$	2.89 ± 8	$0.0064 \pm 2.5 \times 10^{-6}$	$0.00564 \pm 1.7 \times 10^{-6}$
	Heteroscedastic LVGP	0.0115 ± 0.0059	16 ± 22	0.00736 ± 0.0051	0.00814 ± 0.0075
	Separate GP	0.0161 ± 0.012	21.7 ± 24	0.00584 ± 0.0043	0.00746 ± 0.0063
	Categorical kernel	$0.01 \pm 4.3 \times 10^{-8}$	11.4 ± 18	0.00624 ± 0.00086	$0.00564 \pm 3.3 \times 10^{-6}$
EI	Standard LVGP	$0.01 \pm 1.3 \times 10^{-7}$	1.41 ± 0.0072	$0.0064 \pm 8.6 \times 10^{-10}$	$0.00564 \pm 2.3 \times 10^{-7}$
	Heteroscedastic LVGP	$0.01 \pm 9.3 \times 10^{-7}$	20.9 ± 22	0.00572 ± 0.0033	0.00731 ± 0.0062
	Separate GP	0.014 ± 0.01	25 ± 26	0.00524 ± 0.0035	0.00662 ± 0.0045
	Categorical kernel	$0.01 \pm 9.3 \times 10^{-8}$	16.8 ± 21	$0.0064 \pm 5.1 \times 10^{-7}$	$0.00564 \pm 3.5 \times 10^{-6}$
HAEI ($\gamma = 0.5$)	Heteroscedastic LVGP	$0.01 \pm 4.7 \times 10^{-7}$	2.83 ± 7.7	0.00576 ± 0.0016	0.00647 ± 0.0045
	Separate GP	0.014 ± 0.01	21.8 ± 23	0.00524 ± 0.0035	0.00814 ± 0.0075
	Categorical kernel	$0.01 \pm 7.3 \times 10^{-8}$	7.1 ± 15	0.00576 ± 0.0016	$0.00564 \pm 2.8 \times 10^{-6}$
ANPEI ($\beta = 0.8$)	Heteroscedastic LVGP	$0.01 \pm 3.2 \times 10^{-7}$	1.41 ± 0.013	0.00496 ± 0.0022	$0.00564 \pm 1.3 \times 10^{-6}$
	Separate GP	0.011 ± 0.0054	12.1 ± 22	0.00428 ± 0.0037	$0.00564 \pm 2.5 \times 10^{-6}$
	Categorical kernel	$0.01 \pm 9 \times 10^{-8}$	4.64 ± 9.8	0.00445 ± 0.0037	$0.00564 \pm 2.7 \times 10^{-6}$
RAHBO ($\alpha = 0.5$)	Heteroscedastic LVGP	$0.01 \pm 5.5 \times 10^{-7}$	1.41 ± 0.0073	0.00576 ± 0.0016	0.00647 ± 0.0045
	Separate GP	0.017 ± 0.013	3.41 ± 9.6	0.00788 ± 0.0068	0.0102 ± 0.0092
	Categorical kernel	0.0131 ± 0.009	2.34 ± 4.7	0.00563 ± 0.0033	$0.00564 \pm 2.5 \times 10^{-6}$

Table 8: Final noise σ^2 at the incumbent design for every acquisition $\times$ surrogate model ($n_{\text{rep}} = 10$, mean $\pm$ sd over seeds).

Acquisition	Model	Golinski	OTL Circuit	Piston	Rastrigin 6-D
LCB	Standard LVGP	$576 \pm 3.9 \times 10^{-8}$	$0.000194 \pm 1.4 \times 10^{-10}$	$0.000256 \pm 7.3 \times 10^{-10}$	1.97 ± 1.3
	Heteroscedastic LVGP	$576 \pm 5.2 \times 10^{-10}$	$0.000194 \pm 2.7 \times 10^{-11}$	$0.000256 \pm 1.7 \times 10^{-9}$	1.24 ± 0.13
	Separate GP	576 ± 0	0.000651 ± 0.00091	$0.000195 \pm 5.7 \times 10^{-5}$	2.11 ± 2
	Categorical kernel	576 ± 0	0.000194 ± 0	$0.000234 \pm 5 \times 10^{-6}$	1.15 ± 0.079
PI	Standard LVGP	605 ± 0	0.00121 ± 0.00064	$0.000211 \pm 3.9 \times 10^{-5}$	1.16 ± 0.044
	Heteroscedastic LVGP	576 ± 0.0015	0.000474 ± 0.00034	$0.000256 \pm 5.3 \times 10^{-10}$	1.85 ± 1.5
	Separate GP	576 ± 0	0.000194 ± 0	$0.000241 \pm 8.8 \times 10^{-6}$	1.14 ± 0.054
	Categorical kernel	576 ± 0	0.000333 ± 0.00028	$0.000197 \pm 5.6 \times 10^{-5}$	1.11 ± 0.04
EI	Standard LVGP	602 ± 6.2	$0.000886 \pm 3.6 \times 10^{-6}$	$0.000159 \pm 5.3 \times 10^{-5}$	2.97 ± 2
	Heteroscedastic LVGP	$576 \pm 3.9 \times 10^{-6}$	0.000474 ± 0.00034	$0.000233 \pm 4.7 \times 10^{-5}$	1.28 ± 0.16
	Separate GP	576 ± 0	0.000194 ± 0	$0.000241 \pm 8.8 \times 10^{-6}$	1.97 ± 1.7
	Categorical kernel	576 ± 0	$0.000235 \pm 8.1 \times 10^{-5}$	$0.000203 \pm 3.6 \times 10^{-5}$	1.12 ± 0.043
HAEI ($\gamma = 0.5$)	Heteroscedastic LVGP	$576 \pm 7.1 \times 10^{-6}$	0.000474 ± 0.00034	$0.000124 \pm 5.3 \times 10^{-5}$	1.3 ± 0.15
	Separate GP	576 ± 0	0.000194 ± 0	$0.000225 \pm 3.5 \times 10^{-5}$	1.07 ± 0.077
	Categorical kernel	576 ± 0	0.000413 ± 0.00025	$0.000142 \pm 6.1 \times 10^{-5}$	1.09 ± 0.041
ANPEI ($\beta = 0.8$)	Heteroscedastic LVGP	$576 \pm 5.7 \times 10^{-6}$	0.000375 ± 0.00027	$0.000175 \pm 7.4 \times 10^{-5}$	1.12 ± 0.12
	Separate GP	576 ± 0	$0.000235 \pm 8.1 \times 10^{-5}$	$0.000199 \pm 4.4 \times 10^{-5}$	1.17 ± 0.11
	Categorical kernel	576 ± 0	0.000377 ± 0.00027	$0.000137 \pm 2.7 \times 10^{-5}$	1.1 ± 0.031
RAHBO ($\alpha = 0.5$)	Heteroscedastic LVGP	576 ± 0.74	$0.000194 \pm 1.1 \times 10^{-11}$	$0.000256 \pm 6.6 \times 10^{-10}$	1.26 ± 0.073
	Separate GP	589 ± 25	0.000651 ± 0.00091	$0.000178 \pm 5.2 \times 10^{-5}$	1.9 ± 1.4
	Categorical kernel	576 ± 0	0.000194 ± 0	$0.000236 \pm 6.2 \times 10^{-6}$	1.11 ± 0.044

Table 9: Final noise σ^2 at the incumbent design for every acquisition $\times$ surrogate model ($n_{\text{rep}} = 3$, mean $\pm$ sd over seeds).

Acquisition	Model	Ackley 2-D	Branin	Griewank 2-D	Six-hump Camel
LCB	Standard LVGP	$0.01 \pm 5.3 \times 10^{-8}$	23.8 ± 22	0.00608 ± 0.0012	0.00731 ± 0.0062
	Heteroscedastic LVGP	$0.01 \pm 9 \times 10^{-7}$	26.6 ± 22	0.00576 ± 0.0016	0.00731 ± 0.0062
	Separate GP	0.016 ± 0.012	23.7 ± 27	0.00892 ± 0.0073	0.0122 ± 0.015
	Categorical kernel	0.0111 ± 0.0054	10.8 ± 19	0.00608 ± 0.0012	0.00647 ± 0.0045
PI	Standard LVGP	$0.01 \pm 2.2 \times 10^{-7}$	12 ± 19	$0.0064 \pm 2.3 \times 10^{-6}$	$0.00564 \pm 1.8 \times 10^{-6}$
	Heteroscedastic LVGP	0.0118 ± 0.0059	24.7 ± 24	0.0066 ± 0.0048	0.00731 ± 0.0062
	Separate GP	0.011 ± 0.0054	12.9 ± 19	0.00556 ± 0.0034	0.00663 ± 0.0045
	Categorical kernel	$0.01 \pm 1.7 \times 10^{-7}$	15.9 ± 21	$0.0064 \pm 3.9 \times 10^{-6}$	$0.00564 \pm 5.3 \times 10^{-6}$
EI	Standard LVGP	$0.01 \pm 1.5 \times 10^{-7}$	10.4 ± 18	$0.0064 \pm 7.5 \times 10^{-9}$	$0.00564 \pm 4.8 \times 10^{-7}$
	Heteroscedastic LVGP	$0.01 \pm 5.4 \times 10^{-7}$	23.4 ± 22	0.00592 ± 0.0014	0.00731 ± 0.0062
	Separate GP	0.012 ± 0.0075	14.2 ± 18	0.00524 ± 0.0035	0.00731 ± 0.0062
	Categorical kernel	$0.01 \pm 1.8 \times 10^{-7}$	17.3 ± 21	$0.0064 \pm 1.2 \times 10^{-6}$	$0.00564 \pm 3.5 \times 10^{-6}$
HAEI ($\gamma = 0.5$)	Heteroscedastic LVGP	$0.01 \pm 2.7 \times 10^{-7}$	25 ± 22	0.00608 ± 0.0012	$0.00564 \pm 4.8 \times 10^{-6}$
	Separate GP	0.012 ± 0.0075	16.4 ± 21	0.0054 ± 0.0034	0.00662 ± 0.0045
	Categorical kernel	$0.01 \pm 1.5 \times 10^{-7}$	14.9 ± 21	$0.0064 \pm 3.1 \times 10^{-6}$	$0.00564 \pm 5.1 \times 10^{-6}$
ANPEI ($\beta = 0.8$)	Heteroscedastic LVGP	$0.01 \pm 4.8 \times 10^{-7}$	15.1 ± 21	0.00544 ± 0.0019	$0.00564 \pm 2.3 \times 10^{-6}$
	Separate GP	$0.01 \pm 6.2 \times 10^{-7}$	13.6 ± 21	0.00505 ± 0.0045	0.00731 ± 0.0062
	Categorical kernel	$0.01 \pm 6 \times 10^{-7}$	9.12 ± 17	0.00465 ± 0.0023	$0.00564 \pm 7.5 \times 10^{-6}$
RAHBO ($\alpha = 0.5$)	Heteroscedastic LVGP	$0.01 \pm 5.1 \times 10^{-7}$	1.41 ± 0.0085	0.00576 ± 0.0016	0.00647 ± 0.0045
	Separate GP	0.017 ± 0.013	7.48 ± 18	0.00839 ± 0.0071	0.0131 ± 0.016
	Categorical kernel	0.011 ± 0.0054	4.5 ± 12	0.00608 ± 0.0012	$0.00564 \pm 5 \times 10^{-6}$

Table 10: Final noise σ^2 at the incumbent design for every acquisition $\times$ surrogate model ($n_{\text{rep}} = 3$, mean $\pm$ sd over seeds).

Acquisition	Model	Golinski	OTL Circuit	Piston	Rastrigin 6-D
LCB	Standard LVGP	$576 \pm 3.9 \times 10^{-8}$	$0.000194 \pm 5.3 \times 10^{-11}$	$0.000247 \pm 7.7 \times 10^{-6}$	1.28 ± 0.12
	Heteroscedastic LVGP	$576 \pm 4.9 \times 10^{-9}$	$0.000194 \pm 1.7 \times 10^{-11}$	$0.000256 \pm 7.3 \times 10^{-9}$	1.29 ± 0.12
	Separate GP	576 ± 0	0.000651 ± 0.00091	$0.000183 \pm 5.8 \times 10^{-5}$	2.68 ± 1.9
	Categorical kernel	576 ± 0	0.000194 ± 0	$0.000236 \pm 6.2 \times 10^{-6}$	1.13 ± 0.044
PI	Standard LVGP	597 ± 19	0.000753 ± 0.00028	$0.000196 \pm 5 \times 10^{-5}$	1.13 ± 0.089
	Heteroscedastic LVGP	$576 \pm 4 \times 10^{-5}$	0.000514 ± 0.00032	$0.000256 \pm 4.3 \times 10^{-9}$	1.26 ± 0.1
	Separate GP	576 ± 0	0.000194 ± 0	$0.00021 \pm 4.2 \times 10^{-5}$	1.87 ± 1.5
	Categorical kernel	576 ± 0	0.000908 ± 0.0008	$0.000202 \pm 6.4 \times 10^{-5}$	1.15 ± 0.091
EI	Standard LVGP	584 ± 11	0.00079 ± 0.0002	$0.000175 \pm 3.7 \times 10^{-5}$	1.95 ± 1.5
	Heteroscedastic LVGP	$576 \pm 4 \times 10^{-6}$	0.000514 ± 0.00032	$0.000223 \pm 5.7 \times 10^{-5}$	1.24 ± 0.089
	Separate GP	576 ± 0	$0.000275 \pm 9.9 \times 10^{-5}$	$0.000209 \pm 5.3 \times 10^{-5}$	2.49 ± 1.7
	Categorical kernel	576 ± 0	0.000514 ± 0.00032	$0.000173 \pm 6.4 \times 10^{-5}$	1.11 ± 0.037
HAEI ($\gamma = 0.5$)	Heteroscedastic LVGP	$576 \pm 9.1 \times 10^{-6}$	0.000512 ± 0.00032	$0.000185 \pm 7.3 \times 10^{-5}$	1.3 ± 0.21
	Separate GP	576 ± 0	$0.000235 \pm 8.1 \times 10^{-5}$	$0.000211 \pm 3.7 \times 10^{-5}$	1.19 ± 0.1
	Categorical kernel	576 ± 0	0.000275 ± 0.0001	$0.000185 \pm 6.9 \times 10^{-5}$	1.11 ± 0.054
ANPEI ($\beta = 0.8$)	Heteroscedastic LVGP	576 ± 0.00033	$0.000275 \pm 9.9 \times 10^{-5}$	$0.00015 \pm 6.6 \times 10^{-5}$	1.09 ± 0.041
	Separate GP	576 ± 0	$0.000275 \pm 9.9 \times 10^{-5}$	$0.000161 \pm 7.4 \times 10^{-5}$	1.11 ± 0.048
	Categorical kernel	576 ± 0	0.000333 ± 0.00028	$0.000153 \pm 5.9 \times 10^{-5}$	1.1 ± 0.027
RAHBO ($\alpha = 0.5$)	Heteroscedastic LVGP	$576 \pm 2.9 \times 10^{-6}$	$0.000194 \pm 2.8 \times 10^{-11}$	$0.000256 \pm 2.4 \times 10^{-8}$	1.25 ± 0.1
	Separate GP	592 ± 26	0.000651 ± 0.00091	$0.000198 \pm 5.8 \times 10^{-5}$	2.69 ± 1.9
	Categorical kernel	587 ± 22	0.000194 ± 0	$0.000231 \pm 3.8 \times 10^{-5}$	1.18 ± 0.088

Findings. (1) On otl.circuit all models reach top-tier f , but ranked against the whole design space the heteroscedastic LVGP’s noise-aware recommendations come from the **quietest 11%** of designs vs. 34% for both python GPs (percentile framing, `compare.ipynb` Part 6) — equal performance, $\sim 3\times$ quieter design. (2) On rastrigin the incumbent noise ordering is categorical $1.11 < \text{separate } 1.15 < \text{heter } 1.28$ — where the heter LVGP’s f -surface is weak, its noise advantage does not compensate. (3) On the 1-D problems the noise columns mostly mirror which minimum was found (quiet vs noisy twin) — the discriminating robustness view is §6.

6 Heatmaps — acquisition $\times$ model (multi-dimensional problems)

Findings. Reading the four panels column-wise: the Standard LVGP column is the weakest on golinski (noise-blindness, cf. §3) and rastrigin; the heteroscedastic LVGP column is strong on golinski/otl/piston across acquisitions; the categorical GP dominates rastrigin under every acquisition. Acquisition-wise, EI/LCB are the most reliable columns overall; PI is the most erratic.

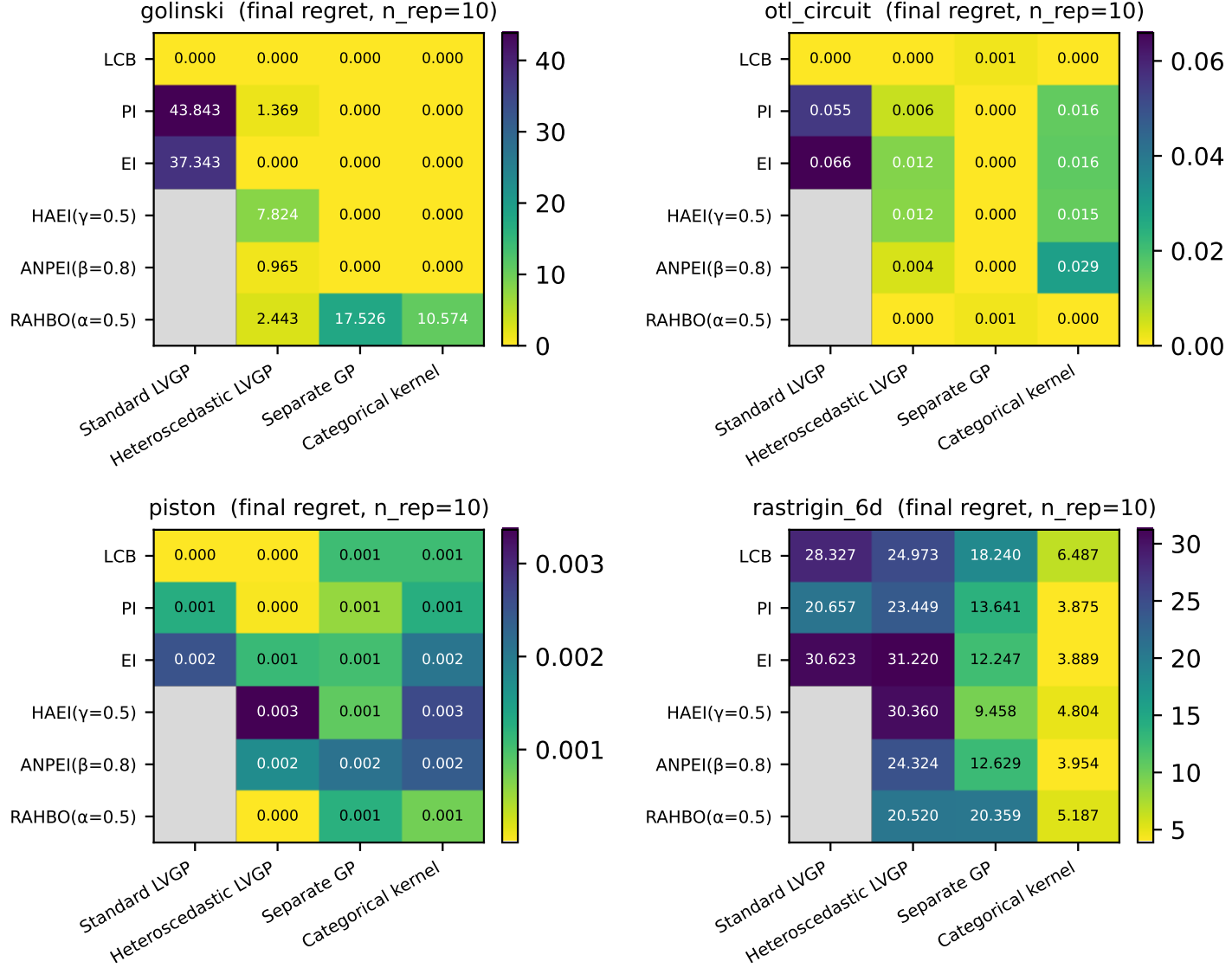


Figure 3: Final regret per acquisition $\times$ model for the multi-dimensional problems ($n_{\text{rep}} = 10$) — the regime where regret is informative. Per-problem versions for the 1-D problems: `compare.ipynb` Part 4.

7 Work in progress — mean–variance (MV) robustness

Robustness is scored on the true $MV_\alpha = f + \alpha\sigma^2$ of the recommended design. Ground truth first: the MV optimum *relocates to a different categorical level* on *branin* ($\alpha \gtrsim 7$), *griewank.2d* ($\alpha \gtrsim 13$) and *golinski* ($\alpha \gtrsim 5$) — the “trap” problems; *camel* and *ackley* are controls whose nominal optima are already quiet. The current sweep runs RAHBO at $\alpha = 0.5$ (below every threshold), so the $\alpha \in \{1, 5, 15\}$ sensitivity sweep and a reporting-rule refinement ([notes/README.md](#)) are pending before conclusions; tables below use $\alpha = 1$.

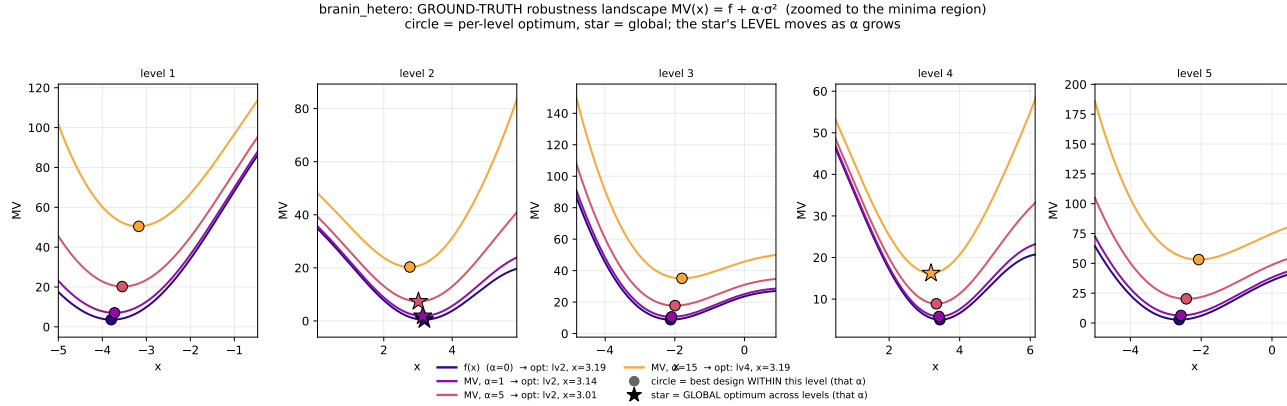


Figure 4: Ground-truth MV landscape, **branin** (trap): the global optimum (star) moves from level 2 to level 4 as α grows.

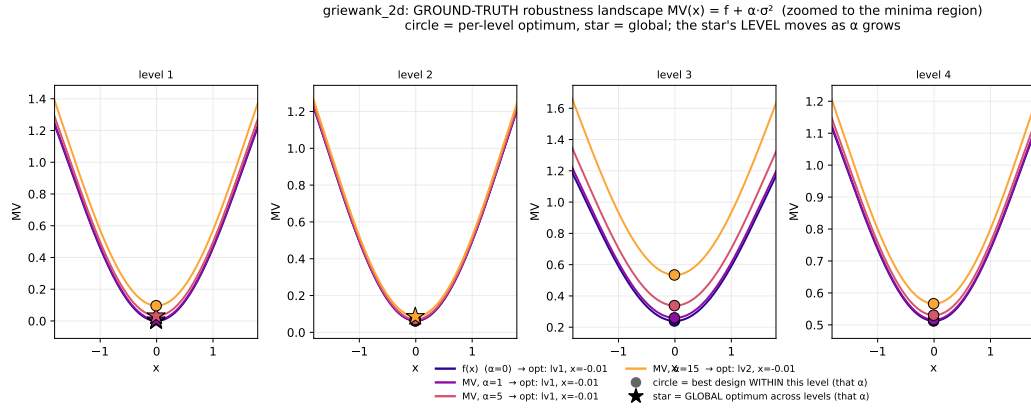


Figure 5: Ground-truth MV landscape, **griewank_2d** (trap at $\alpha \gtrsim 13$: level 1 $\rightarrow$ 2).

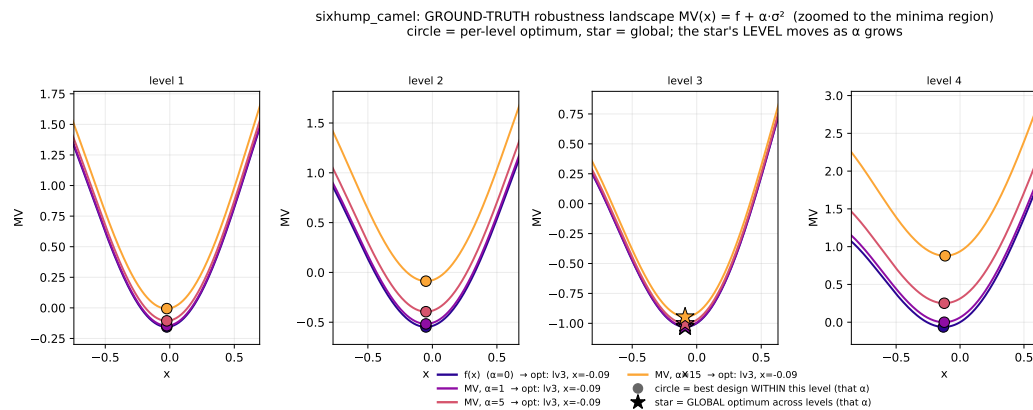


Figure 6: Ground-truth MV landscape, **sixhump_camel** (control: the optimum never moves — already quiet).

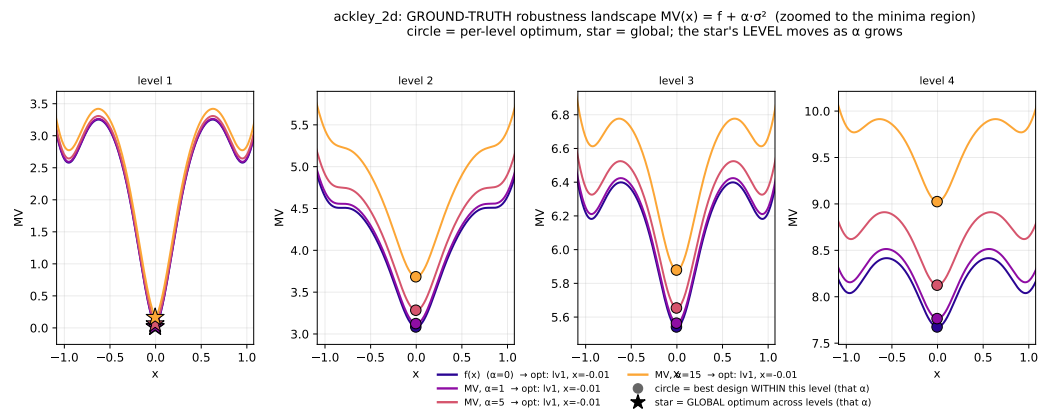


Figure 7: Ground-truth MV landscape, **ackley_2d** (control).

Findings. (1) **The branin trap catches the categorical-kernel GP:** a cluster of its noise-aware recommendations sits on the noisy twin ($\sigma^2 \approx 40\text{--}80$, dominated), while the heteroscedastic LVGP’s recommendations sit on the Pareto front near the quiet optimum. (2) Only 3 of the 8 completed problems (branin, griewank_2d, golinski) can discriminate robust from non-robust methods; camel/ackley confirm no method is penalized when robustness is free. (3) At the sweep’s $\alpha = 0.5$ the robust and nominal optima coincide everywhere, so the MV tables below ($\alpha = 1$) largely mirror the regret tables; the discriminating comparison is the pending $\alpha \in \{5, 15\}$ sweep.

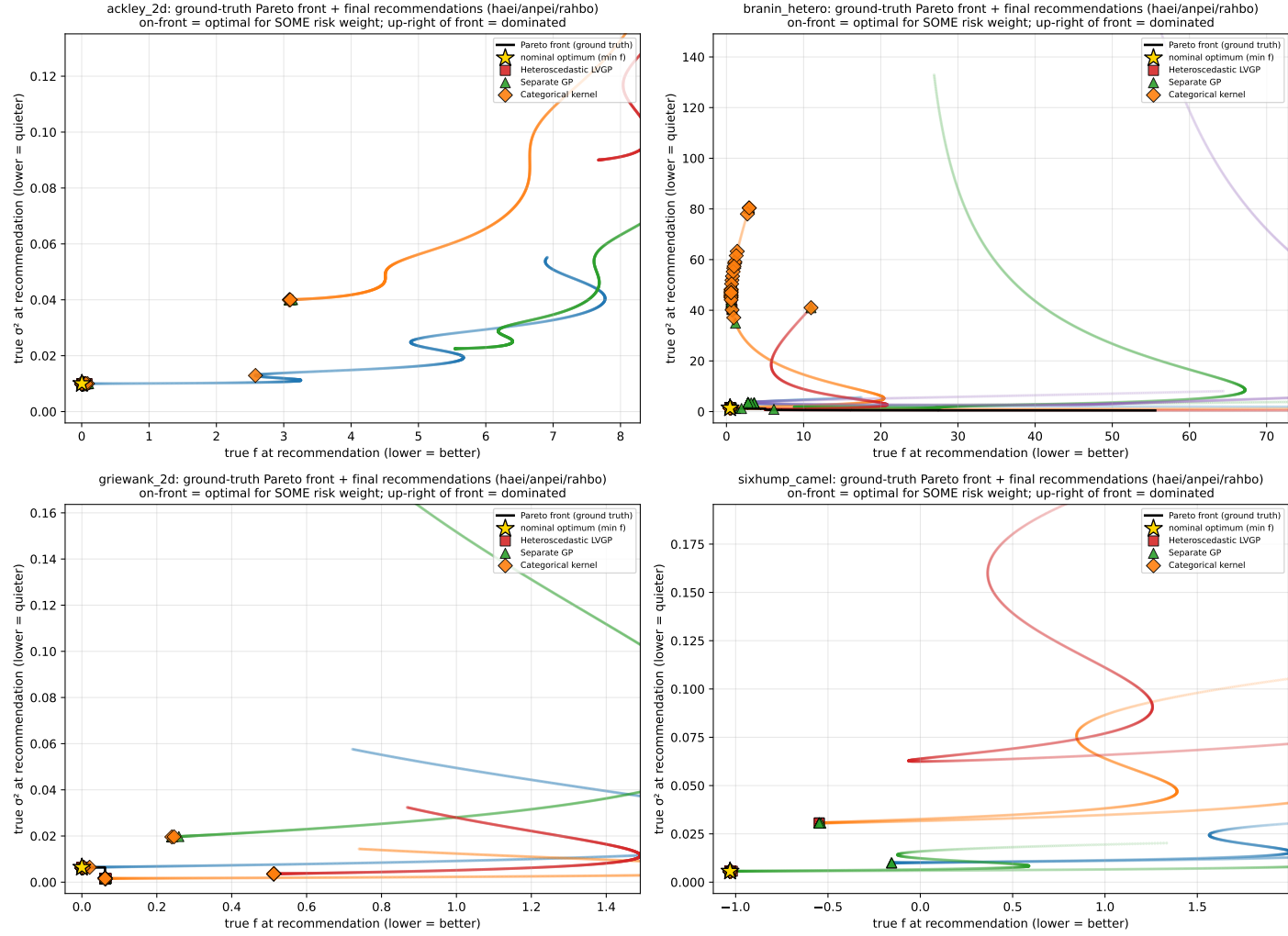


Figure 8: Hyperparameter-free view: ground-truth (f, σ^2) Pareto front per 1-D problem, with each model's final noise-aware recommendations overlaid. On-front = optimal for some risk weight; up-right = dominated.

Table 11: Final mean-variance regret (RAHBO robustness, $MV = f + \alpha\sigma^2$, $\alpha = 1$) for every acquisition $\times$ surrogate model ($n_{\text{rep}} = 10$, mean $\pm$ sd over seeds).

Acquisition	Model	Ackley 2-D	Branin	Griewank 2-D	Six-hump Camel
LCB	Standard LVGP	0.00408 $\pm$ 0.0047	0.00407 $\pm$ 0.0032	0.00381 $\pm$ 0.014	0.0339 $\pm$ 0.13
	Heteroscedastic LVGP	0.0188 $\pm$ 0.021	1 $\pm$ 1.6	0.0097 $\pm$ 0.021	0.034 $\pm$ 0.13
	Separate GP	0.733 $\pm$ 1.3	2.34 $\pm$ 3.4	0.0784 $\pm$ 0.1	0.207 $\pm$ 0.34
	Categorical kernel	0.297 $\pm$ 0.88	1.77 $\pm$ 4.3	0.019 $\pm$ 0.064	0.0193 $\pm$ 0.1
PI	Standard LVGP	0.0131 $\pm$ 0.014	0.139 $\pm$ 0.74	0.000902 $\pm$ 0.0012	0.000115 $\pm$ 0.00017
	Heteroscedastic LVGP	0.281 $\pm$ 1	2.41 $\pm$ 4.5	0.0436 $\pm$ 0.085	0.0513 $\pm$ 0.15
	Separate GP	0.715 $\pm$ 1.3	1.72 $\pm$ 2.2	0.0343 $\pm$ 0.064	0.0702 $\pm$ 0.2
	Categorical kernel	0.00491 $\pm$ 0.006	0.854 $\pm$ 3.3	0.00234 $\pm$ 0.01	0.000646 $\pm$ 0.0023
EI	Standard LVGP	0.0102 $\pm$ 0.0098	0.00449 $\pm$ 0.0035	$-6.55 \times 10^{-7} \pm 4.2 \times 10^{-7}$	$1.67 \times 10^{-6} \pm 5.6 \times 10^{-6}$
	Heteroscedastic LVGP	0.022 $\pm$ 0.031	0.868 $\pm$ 1.7	0.0218 $\pm$ 0.049	0.0341 $\pm$ 0.13
	Separate GP	0.426 $\pm$ 1.1	1.81 $\pm$ 2	0.0278 $\pm$ 0.05	0.0467 $\pm$ 0.18
	Categorical kernel	0.01 $\pm$ 0.0092	0.765 $\pm$ 1.3	0.000174 $\pm$ 0.00025	0.000521 $\pm$ 0.00094
HAEI ($\gamma = 0.5$)	Heteroscedastic LVGP	0.0259 $\pm$ 0.023	0.17 $\pm$ 0.79	0.00804 $\pm$ 0.019	0.0171 $\pm$ 0.091
	Separate GP	0.429 $\pm$ 1.1	1.29 $\pm$ 1.8	0.0277 $\pm$ 0.05	0.0511 $\pm$ 0.15
	Categorical kernel	0.00801 $\pm$ 0.0085	0.0895 $\pm$ 0.19	0.00886 $\pm$ 0.02	0.000347 $\pm$ 0.00085
ANPEI ($\beta = 0.8$)	Heteroscedastic LVGP	0.0208 $\pm$ 0.019	0.00777 $\pm$ 0.022	0.018 $\pm$ 0.026	$6.35 \times 10^{-5} \pm 0.00015$
	Separate GP	0.117 $\pm$ 0.56	1.42 $\pm$ 2.1	0.0392 $\pm$ 0.049	0.000237 $\pm$ 0.00042
	Categorical kernel	0.00552 $\pm$ 0.0094	0.159 $\pm$ 0.47	0.0387 $\pm$ 0.05	0.000346 $\pm$ 0.00088
RAHBO ($\alpha = 0.5$)	Heteroscedastic LVGP	0.0203 $\pm$ 0.025	0.00399 $\pm$ 0.012	0.00786 $\pm$ 0.019	0.017 $\pm$ 0.091
	Separate GP	0.73 $\pm$ 1.3	0.581 $\pm$ 1.5	0.0783 $\pm$ 0.1	0.173 $\pm$ 0.3
	Categorical kernel	0.4 $\pm$ 1	0.717 $\pm$ 3.1	0.0403 $\pm$ 0.1	0.00022 $\pm$ 0.00055

Table 12: Final mean-variance regret (RAHBO robustness, $MV = f + \alpha\sigma^2$, $\alpha = 1$) for every acquisition $\times$ surrogate model ($n_{\text{rep}} = 10$, mean $\pm$ sd over seeds).

Acquisition	Model	Golinski	OTL Circuit	Piston	Rastrigin 6-D
LCB	Standard LVGP	$3.48 \times 10^{-6} \pm 1.1 \times 10^{-7}$	$3.87 \times 10^{-6} \pm 3.2 \times 10^{-6}$	$2.16 \times 10^{-7} \pm 1.5 \times 10^{-7}$	29.2 ± 7.3
	Heteroscedastic LVGP	$1.72 \times 10^{-7} \pm 1.8 \times 10^{-9}$	$4.64 \times 10^{-7} \pm 3.8 \times 10^{-7}$	$4.6 \times 10^{-7} \pm 5.1 \times 10^{-7}$	25.1 ± 5.7
	Separate GP	0 ± 0	0.00188 ± 0.0038	0.00098 ± 0.00066	19.2 ± 11
	Categorical kernel	0 ± 0	0 ± 0	0.00102 ± 0.00023	6.53 ± 4.3
PI	Standard LVGP	72.5 ± 0	0.056 ± 0.029	0.00122 ± 0.00051	20.7 ± 6.8
	Heteroscedastic LVGP	1.37 ± 2.6	0.00586 ± 0.0099	$2.93 \times 10^{-5} \pm 3.7 \times 10^{-5}$	23.5 ± 3.1
	Separate GP	0 ± 0	$6.94 \times 10^{-5} \pm 7 \times 10^{-5}$	0.000517 ± 0.00037	13.7 ± 3.8
	Categorical kernel	0 ± 0	0.0159 ± 0.011	0.00119 ± 0.0011	3.88 ± 2.8
EI	Standard LVGP	62.9 ± 19	0.0666 ± 0.03	0.0024 ± 0.002	32.5 ± 11
	Heteroscedastic LVGP	$6.77 \times 10^{-5} \pm 8.2 \times 10^{-5}$	0.0124 ± 0.014	0.00106 ± 0.00057	31.4 ± 2.6
	Separate GP	0 ± 0	$3.07 \times 10^{-5} \pm 6.1 \times 10^{-5}$	0.00099 ± 0.00074	13.1 ± 9.6
	Categorical kernel	0 ± 0	0.0163 ± 0.014	0.002 ± 0.00056	3.91 ± 2.2
HAEI ($\gamma = 0.5$)	Heteroscedastic LVGP	7.82 ± 4.1	0.0119 ± 0.013	0.00323 ± 0.0018	30.6 ± 5
	Separate GP	0 ± 0	$3.07 \times 10^{-5} \pm 6.1 \times 10^{-5}$	0.000791 ± 0.00055	9.43 ± 2.5
	Categorical kernel	0 ± 0	0.0157 ± 0.013	0.00249 ± 0.00091	4.79 ± 1.6
ANPEI ($\beta = 0.8$)	Heteroscedastic LVGP	0.965 ± 1.4	0.00408 ± 0.0075	0.00165 ± 0.0013	24.3 ± 5.9
	Separate GP	0 ± 0	0.000202 ± 0.00037	0.00203 ± 0.00067	12.7 ± 3.5
	Categorical kernel	0 ± 0	0.0289 ± 0.017	0.00226 ± 0.00077	3.96 ± 2
RAHBO ($\alpha = 0.5$)	Heteroscedastic LVGP	2.72 ± 2.4	$2.74 \times 10^{-7} \pm 1.7 \times 10^{-7}$	$4.57 \times 10^{-7} \pm 5.5 \times 10^{-7}$	20.7 ± 4.5
	Separate GP	28.8 ± 51	0.00188 ± 0.0038	0.00115 ± 0.00057	21.2 ± 8.6
	Categorical kernel	10.6 ± 4	0 ± 0	0.000695 ± 0.00033	5.2 ± 2.6

Pointers

- `notebooks/compare.ipynb` — full analysis (9 parts): convergence facets, per-problem reports, latent spaces, acquisition tables, robustness visuals (MV/quantile landscapes, Pareto, percentiles), model-quality metrics, exports.
- `notebooks/compare_v2.ipynb` — curated narrative with equations and dated observation notes.
- `notebooks/compare_matlab_python.ipynb` — MATLAB vs native-python LVGP validation (equivalence, computing time, latent match).